

SHUWEN GONG

System Modelling & Control · Vehicle Dynamics & Powertrain

PROFILE

Mobility Engineering M.Sc. at Chalmers, focused on vehicle system modelling and control in MATLAB/Simulink and on identifying models from measured data. Available for a 30 ECTS thesis from January 2027. Earlier: fighter-aircraft final assembly (3 yrs) and live-service game systems design (4.5 yrs) — the habit of looking at the whole system before changing a part.

CONTACTS

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PORTFOLIO

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SKILLS

Control & Identification
State feedback, pole placement
Observer design, discretisation
System identification
Signal processing

Modelling & Simulation
MATLAB, Simulink, QSS
Vehicle dynamics, quarter-car
Hybrid powertrains, driving cycles

Data & Software
IMU data, PSD, filtering
Python, Git

Systems & Process
PFMEA, PDM/ERP/CAPP, CATIA

Languages
English (professional), Chinese (native), Swedish (learning)

EDUCATION

M.Sc. Mobility Engineering — Sep 2025 – Jun 2027 (expected)
Chalmers University of Technology — Gothenburg, Sweden

- Coursework: Systems and Mechatronics for Mobility Engineering; Electric and Hybrid Vehicles; Propulsion and Energy Systems for Transport; Vehicle and Traffic Safety; Connected Fleets in Data-Driven Engineering.
- Reversing aid for a tractor–semitrailer: pole-placement state feedback with a discretised observer, validated on the full nonlinear model across trailer lengths and speeds (Simulink).
- Series and parallel HEV powertrain design for a Class M reference vehicle: component sizing from top-speed and acceleration requirements, battery state-of-charge management, engine operating-point placement over NEDC and a custom cycle (QSS).
- Ride comfort on a Volvo 7900 Electric: bus IMU data → road-excitation PSD → quarter-car model; suspension stiffness and damping optimised to cut body-acceleration RMS under deflection and tyre-force constraints (MATLAB).

B.Eng. Aircraft Manufacturing Engineering — Sep 2013 – Jun 2017
Nanjing University of Aeronautics and Astronautics (NUAA) — Nanjing, China

- Coursework in dynamics, control fundamentals, and engineering mathematics.

PROJECTS & AWARDS

Winner, Gothenburg Tech Week × Chalmers Hackathon 2026 — Sep 2026
Team Go Pikachu, SKF challenge on training industrial humanoids Gothenburg, Sweden

- Composed pretrained locomotion skills for a Unitree G1 in MuJoCo, with a height scan and a factory map choosing the skill, an ISO/TS 15066 safety layer, and shared simulation test gates; fine-tuned only on the failures the gates exposed — 99 % on a 220-run terrain matrix, 88 % on a 19.9 m factory route with stairs.

WORK EXPERIENCE

Production Engineer, Aircraft Final Assembly — Jul 2017 – May 2020
AVIC Chengdu Aircraft Industrial (Group) Co., Ltd. — Chengdu, China

- Process planning and PFMEA for fighter-aircraft final-assembly stages; coordinated mechanical, avionics and quality teams across multi-stage integration. Led on-site technical support in Pakistan for a new aircraft model deployment.
- First inventor, granted utility-model patent CN 210090631 U: a portable weight-on-wheels switch inspection device for a flight-critical final-assembly check.

Game Systems Designer — Aug 2020 – Feb 2025
Aoshitang · Zlong Games · Duoyi Network — China

- Owned economy and progression rule systems, feature roadmaps and live-ops planning for four shipped live-service games reaching hundreds of thousands of players.
- Defined KPIs and tuned rule systems against player behaviour data after each release.